Vandan Agrawal

vandanagrawal1203@gmail.com | +1 (352)-721-8261 | linkedin.com/in/vandanagrawal | Projects

Education

University of Florida **December 2025**
Master of Science in Computer Science | GPA: 3.66
Amity University **May 2024**
Bachelor of Technology in Computer Science and Engineering | GPA: 3.18

Skills

- **Languages:** Python, Java, C++, C#, SQL, Apex, C, Golang
- **Frameworks and Libraries:** Tkinter, NumPy, TensorFlow, Scikit-learn, Salesforce LWC, Pandas
- **Developer tools:** Git, GitHub, AWS, Jupyter Notebook, Google Colab
- **Specialized Systems:** Unity 3D, Blender, Ghidra, x64dbg, Wireshark, Splunk, VMware
- **Operating Systems:** Windows, Linux (Debian, Kali, Ubuntu), MacOS

Work Experience

Cybersecurity Analyst Intern, *Malware Reverse Engineering Lab* **January 2025 – May 2025**
- Reverse-engineered over 10 real-world malware samples analyzing direct system calls in x64 assembly to identify execution flow, persistence, C2, and IOCs, enabling rapid incident response against threats.
- Performed 60+ hours of detailed static and dynamic analysis using Ghidra, Wireshark, x64dbg, etc. on a multi-system sandbox to extract features like execution paths and network indicators.
- Delivered 4 technical reports highlighting obfuscation techniques and behavioral signatures, providing standardized documentation designed to assist incident response teams in threat detection and mitigation
- Dissected a variant of the Akira ransomware as a final project, documenting behavior and IOCs which has impacted over 250 small to medium-sized American organizations since March 2023.
- Authored YARA rules for 5+ malware by documenting behavioral and threat actor signatures, improving endpoint security to detect payload variants and strengthening organizational security postures.

Software Developer Intern, *SmartInternz* **May 2023 – July 2023**
- Customized and virtually deployed over 20 Salesforce objects, workflows, and permission models to enforce secure, role-based access control and streamline business operations using Object manager.
- Engineered asynchronous Apex triggers and batch processes to manage high-concurrency data without exceeding multi-tenant governor limits improving system responsiveness by 15%.
- Automated security compliance monitoring by implementing Shield Platform Encryption and Security Health Check tools, reducing manual data-integrity reviews by 60% for 5000+ sensitive records.
- Orchestrated asynchronous REST integrations and complex Screen Flows that impacted operational efficiency to automate 500+ weekly transactions across 3 cross-functional simulated departments.

Projects

GatorGlide Delivery Co. - Logistics Order Management System - *Python*
- Engineered a dual AVL Tree-based system in Python to synchronize order priorities and ETAs in O(log n).
- Implemented automated state-validation by utilizing O(1) hash-map lookups and conditional logic to compare real-time system timestamps against calculated departure windows for invalid order cancellations.
- Architected an automated validation suite to parse input test data and cross-reference system output against baseline text files to ensure 100% logical accuracy for regression testing cycles.

MEDEX - A Pharmacy Retail Software - *Python, MySQL*
- Developed sales software for a model pharmacy with Python for both front-end and back-end, with MySQL.
- Designed login systems making use of First Day Passwords for secure employee onboarding.
- Implemented customers, employees, inventory, and billing data retention using my-sql-connector.
- Limited access to features depending on employee posts, imitating Role-Based access control.
- Architected backend processes using python with field sanity checks to reduce vulnerability.

Hill Rush - The Game - *Unity 3D, Blender, C#*
- Prototyped and optimized a 3D physics-based game on Unity game engine, writing scripts for game logic such as object spawning and score tracking in C# and created assets for it in Blender.
- Designed core loop of the game which includes placing temporary barriers on a downward inclined slope to prevent constantly falling objects from hitting characters trying to cross a road.

Competitions

- **WRCCDC 2025 –** Blue Team; contributed to service uptime, incident response, network hardening, and inject execution during an active defensive cybersecurity competition.
- **Amicode 2023 –** Winner in an Inter-university competitive programming competition.